

Adarsh Chatlapally

Hyderabad, IN | +91 6304358472 | saiadarsh22@gmail.com | [LinkedIn](#) | [Portfolio](#) | [Chatbot](#)

Skills

Inference Optimizations | C/C++/Python | AI Agents | PyTorch, CUDA | Docker | Linux OS | CMake, Bazel | Git | CNN/GenAI Models | Jenkins

Work Experience

Dheyo AI

Hyderabad, IN

AI Engineer — Skills: Agentic AI, CUDA, PyTorch, Nix

Sep 2025 – Present

- Solely developed and built a GPU kernel framework for writing, testing, and packaging custom fused kernels as PyTorch operators, with an integrated heatmap visualizer for data checks; registered the kernel as a Python package that auto-selects fused/shape-specific kernels to reduce inference speed. Results: QWEN-2512 and Z-Image turbo ran at 1.2x eager mode without `torch.compile()`
- Built a PG-vector database for asset search using HNSW search and a QWEN embedding model, enabling a scene-creation agent to fetch relevant asset files to place in the environment
- Worked with TPU pipelines and patched FlashAttention-style kernels to enable scaled runs
- Built NixOS config flakes to help build repo versions without conflicts and compile them from source
- Created a CUDA agent that uses the GPU kernel framework to automatically write kernel templates for a given specific model
- Benchmarked Image Gen models on H100 GPU's using different parallelism techniques in vLLM with different output size, steps

Kinara AI (/NXP Semiconductors)

Hyderabad, IN

Kernel Developer Intern — Skills: C, Python, ML operators, Tensilica processor

Nov 2024 – May 2025

- Worked extensively with kernels such as convolution, shift, element-wise kernels, and SoftMax for the Ara-2 processor
- Developed kernels for YOLO models with quantized input values (from float32 to int8)
- Identified and added checkers for an instruction bug, integrating the solution into the Jenkins pipeline
- Restructured the kernel library to optimize pull request execution time, reducing it from 3 hours to 10–20 minutes
- Gained hands-on experience with Gitea, Linux systems, and shell commands
- Experienced with the Tensilica processor instruction set

All India Council for Technical Education

Remote

Intern

May 2023 – Jul 2023

- Built a strong foundation in core AI and ML concepts, with a focus on understanding model architectures relevant to GenAI, such as transformers and encoder-decoder structures
- Completed hands-on mini projects to simulate real-world use cases, including model training, evaluation, and basic automation, preparing for more complex GenAI applications

Research Centre Imarat, DRDO

Hyderabad, IN

Intern — Skills: C, Python, TCP/IP, Controller

Jun 2023 – Jul 2023

- Developed a client-server application using TCP/IP socket programming to enable communication over a network
- Implemented a multi-client chatting server to allow multiple clients to connect and exchange messages

Projects

GPU Kernel Framework | [link](#)

- Developed a Python-based framework to streamline the creation, benchmarking, and validation of custom GPU kernels using CUDA and PyTorch
- Engineered features for easy kernel implementation and performance testing, simplifying the development workflow for GPU-accelerated computing tasks

Enhanced School Zone Safety: Vehicle Speed Estimation and Geofence Based Control

- Developed a real-time vehicle speed estimation system using YOLO and Deep SORT, enhancing school zone safety through intelligent geofencing; implemented geofence-based speed control to automatically detect and manage vehicle behavior in school zones, improving pedestrian safety

RAG based Chatbot | [link](#)

- Developed a personal Retrieval-Augmented Generation (RAG) chatbot designed to answer questions based on the content of my resume; uses Neo4j to store and query semantically structured data, enabling accurate context retrieval
- Backend hosted on Hugging Face Spaces leverages sentence transformers for embedding and similarity search; designed a clean, interactive frontend that mimics a typical LLM chat interface, complete with dynamic input placement and generic prompt suggestions to guide users

AI based Cover Letter Generator | [link](#)

- Developed a cover letter generator Google extension that streamlines the creation of downloadable, well-formatted cover letters for job applicants; generates letters based on resumes and job descriptions, and users can provide their own LLM API key

Certifications

Supervised Machine Learning: Regression and Classification (DeepLearning.ai)

AWS Academy Cloud Foundation (Amazon Web Services)

AWS Academy Machine Learning Foundations (Amazon Web Services)

Network Essentials (CISCO)

Education

SRM Institute of Science and Technology <i>Bachelor of Technology, Computer Science and Engineering (AI & ML) — GPA: 8.16/10</i>	Chennai, IN <i>2021 – 2025</i>
Sri Chaitanya Junior College <i>MPC — Percentage: 92.4%</i>	Hyderabad, IN <i>2020 – 2021</i>
Johnson Grammar School <i>Indian Certificate of Secondary Education — Percentage: 85%</i>	Hyderabad, IN <i>2018 – 2019</i>